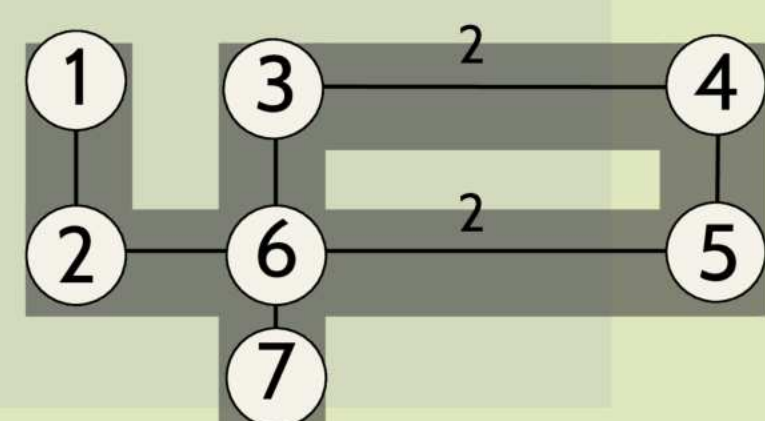


Growth

- Agar plate
- Oats for food
- Continuous subculturing

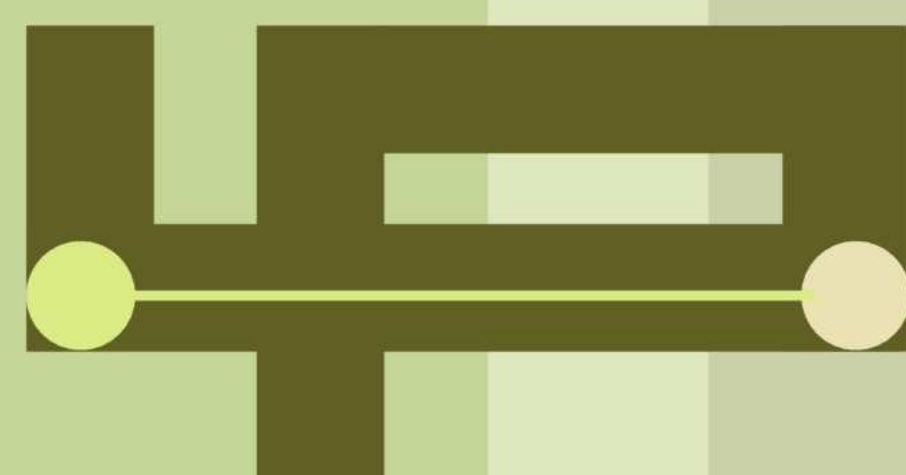
Setup

- 3D printed Academies model
- Mapped on graph as nodes and edges
- Camera



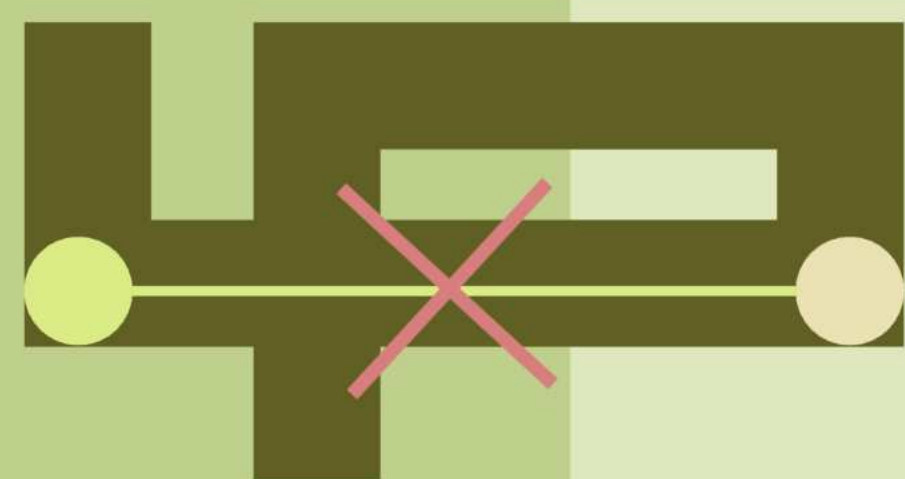
Inoculation

- Set ending points with oats
- Place *P. polycephalum* at starting point



Adaptation

- Block routes with table salt and repeat previous step



Comparison

- A* algorithm (control)
- Kruskal Wallis

Results

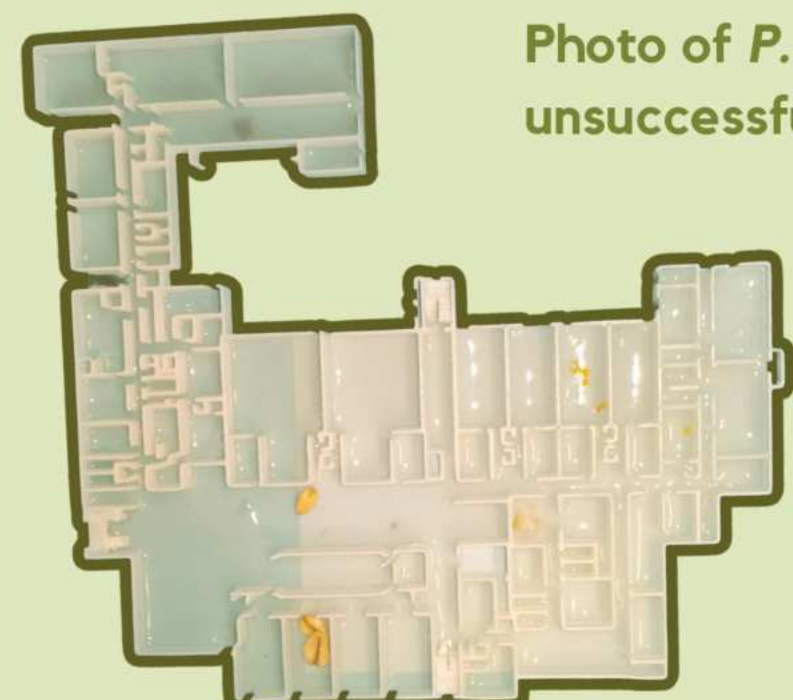


Photo of *P. polycephalum* in maze; unsuccessful due to culture drying out

- Halted data collection
 - Repeated contamination
 - Insufficient environmental conditions

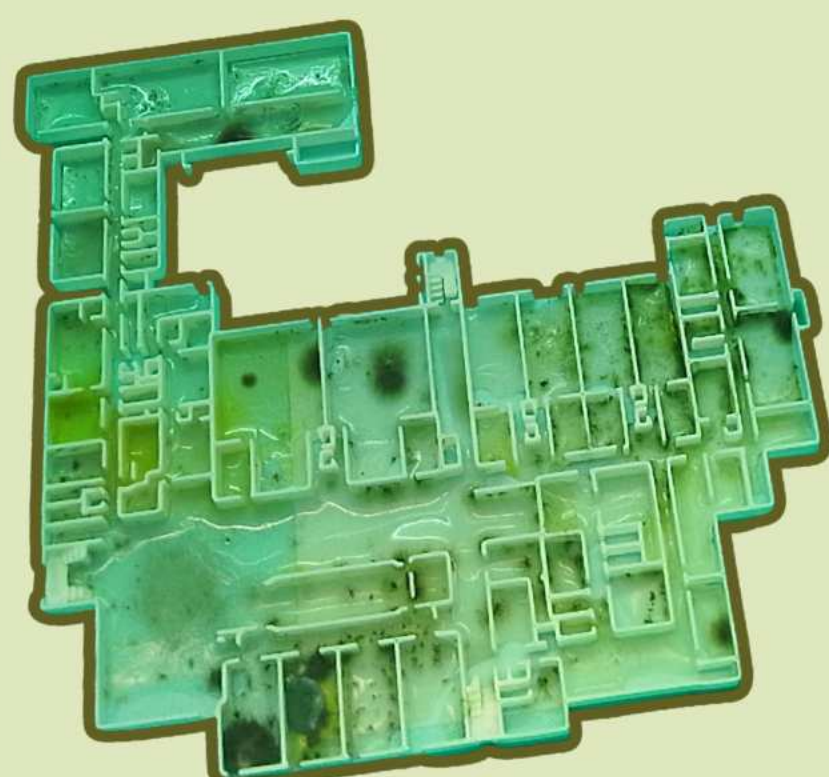


Photo of *P. polycephalum* in maze; unsuccessful due to contamination

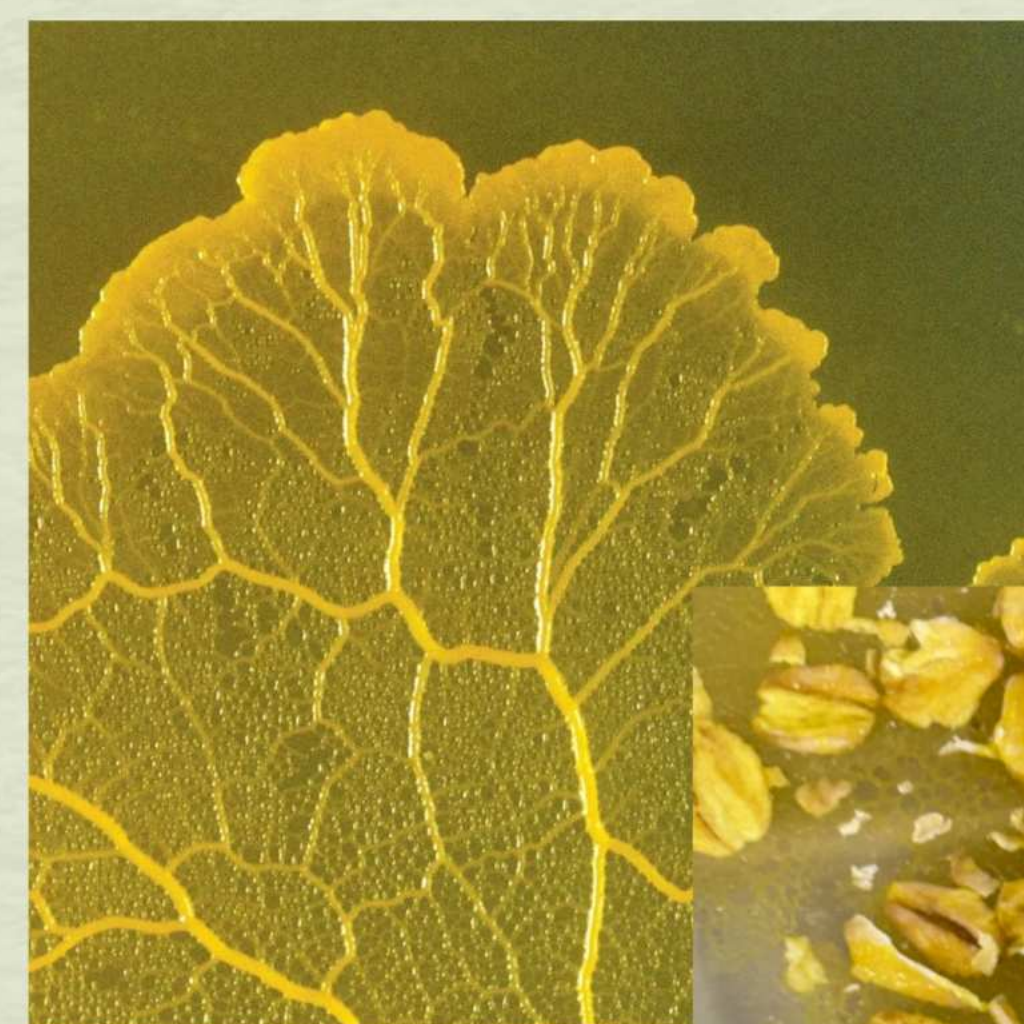
Methodology

Analyzing efficiency of slime mold *Physarum polycephalum* as a model for generating evacuation routes in the Academies of Loudoun building

MAZE

RUNNERS

BACKGROUND INFORMATION



Scott Camazine. (n.d.). *Physarum polycephalum* [Photograph]. Science Source. <https://www.scientificamerican.com/article/how-slime-molds-remember-where-they-ate/>



P. polycephalum on agar plate feeding on oats

PURPOSE:

Address current limitations in current evacuation route plans through the use of a model organism.

IV

START/END POINTS

Controls

- TEMPERATURE (~22°C)
- LIGHT LEVELS (COVERED WITH ALUMINUM FOIL)

Hypothesis

P. polycephalum will be used to create efficient routes with similar efficiency to those created by an A-star (A*) algorithm. This is due to *P. polycephalum*'s cost-effective food foraging abilities, as well as its gravitropic traits that can apply to three dimensions.

P. polycephalum propagating on nylon substrate



Adamatzky, A. (2003). Route 20, Autobahn 7, and Slime Mold: Approximating the Longest Roads in USA and Germany With Slime Mold on 3-D Terrains. [Photograph]. 10.1109/TCYB.2013.2248359

PATH
CREATED BY
SLIME MOLD

DV

Conclusions

- Inconclusive results
 - *P. polycephalum* unable to complete maze due to repeated contamination
- Preliminary trials do not support hypothesis
 - *P. polycephalum* traveled opposite to predicted path
 - Indicates experimental error; contradicts standard studies
- Future experiments may benefit from fully axenic strains and rigorous environmental control
- Successful experimentation could be implemented into path-making algorithms for use in building design

Key References

- Adamatzky, A. (2014). Route 20, Autobahn 7, and slime mold: Approximating the longest roads in USA and Germany with slime mold on 3-D terrains. *IEEE Transactions on Cybernetics*, 44, 126–136.10.1109/TCYB.2013.2248359
- Evangelidis, V., Jones, J., Dourvas, N., Tsompanas, M.-A., Sirakoulis, G.C., Amanatzky, A. (2017). *Physarum machines imitating a Roman road network: The 3D approach. Scientific Reports*, 7. <https://doi.org/10.1038/s41598-017-06961-y>
- Hossain, T., Uddin, S., & Hasan, A. (2021). Comprehensive Review of the A* Algorithm: Theory, Implementation, and Applications. *Independent University*. 10.13140/RG.2.2.29143.04003
- Miguel Ferreira Simões, J. (2023). Building evacuation system. <https://repositorioaberto.up.pt/handle/10216/156561?mode=full>
- Nakagaki, T., Yamada, H. & Tóth, Á. (2000). Maze-solving by an amoeboid organism. *Nature*, 407(470). <https://doi.org/10.1038/35035159>.
- NIST. (2009, March 24). *NIST stairwell evacuation study finds 'what we know we don't know.'* <https://www.nist.gov/news-events/news/2009/03/nist-stairwell-evacuation-study-finds-what-we-know-we-dont-know>
- scientific department. (2025, September 30). *Challenges and Solutions for Emergency Egress Simulation in Historic Buildings. Fire Risk Heritage.* <https://www.fireriskheritage.net/analysis-of-risks-and-solutions-for-cultural-heritage/challenges-and-solutions-for-emergency-egress-simulation-in-historic-buildings/>